

NOTE · The 50 Hinge

THE 50 HINGE

Six Independent Appearances of One Integer Across Scale

D. B. – April 2026 (updated 5 May 2026; 16 September 2026)

Gap Geometry – <https://gap-geometry.github.io/sqrt2-ln2-geometric-constants-/about.html>

OSF: osf.io/zx4g7

► READING UPDATE – 2026-09-16. No value in this paper is changed by this notice.

How the corpus is now read: One coordinate up – <https://osf.io/ujwc7>

DOI 10.17605/OSF.IO/UJWC7

★ UPDATE NOTICE – 16 September 2026 ★

This version adds three lines under the title (a pointer to One coordinate up, where the corpus is now read) and carries one erratum and one note on the count; every number, factorization and deviation stands.

- (1) Three lines added under the title – READING UPDATE – pointing to One coordinate up (<https://osf.io/ujwc7> · DOI 10.17605/OSF.IO/UJWC7), the synthesis where the corpus is now read.
- (2) This paper is classed as a NOTE on the Observational Record shelf of the framework's Direct Documents page, under the short title "The 50 Hinge".
- (3) ERRATUM, §9 (GAP SCALING): "The correction $1/2500 = 1/50^2$ emerges from the continued fraction expansion of ρ " – it does not. The partial quotients of ρ are [36; 2, 1, 3, 20, 2, 1, 384, ...] and its convergents after 400/11 are 8109/223, ...; no 50 and no 2500 occurs. $1/2500$ is the rounded unit fraction chosen at the second step of the gap scaling formula (The 400/11 Formula, §3.2: "remarkably close to $-1/2500$ "). Only 400/11 is a convergent. A correction line is added in place; appearance 4 stands as an arithmetic fact ($2500 = 50^2$) about a chosen denominator.
- (4) THE COUNT: the title's six appearances are not six independent computations. Appearances 2 and 3 are one fact – $1/G = 50.6538\dots$, its floor is the first partial quotient and the nearest step is 50 – as this paper's own §9 already says; and appearance 4 rests on a chosen denominator (item 3). The abstract's "six independent computations" is read with this notice; the title stands as printed.

Every value re-computed 2026-09-11 at 50 digits (both continued fractions to eleven terms, $50 \cdot G$, $82 \cdot G$, $49/50$, the DESI residuals from the quoted values, the correction series): all stand. Prior version remains on OSF in the project's archive/ folder, named by its date.

★ UPDATE NOTICE – 5 May 2026 ★

This version aligns evidence-level labels with the canonical methodology taxonomy and renames the §5 correction-scale labels to avoid a word collision. Both changes are definitional. They do not change the scope, honesty, or calculations of this document. Every numerical claim, factorization, deviation, and falsification criterion is unchanged.

(1) §11 EVIDENCE CLASSIFICATION – LABEL ALIGNMENT

Earlier labels: LEVEL 1 – ALGEBRAIC / EXACT, LEVEL 2 – OBSERVED IN DATA, LEVEL 3 – STRUCTURAL PATTERN. Aligned to the canonical methodology page (gap-geometry.github.io methodology): LEVEL 1 – EXACT [Theorem], LEVEL 2 – EMPIRICAL [Observation], LEVEL 3 – CONJECTURAL [Conjecture]. Item assignments inside each level are unchanged.

(2) §5 CORRECTION HIERARCHY – RENAMED FROM "Level" TO "Order"

The §5 correction hierarchy used "Level 0 / Level 1 / Level 2" as scale-order labels for the gap scaling formula's leading term and its first two corrections. The word "Level" collided with the evidence-level taxonomy used elsewhere in the framework. Renamed to "Order 0 / Order 1 / Order 2" so a reader does not parse "Level 1: 1/2500" as an evidence-level claim.

Prior published version archived as OLD-2026-05-05_50_Hinge_Standalone.txt and OLD-2026-05-05_50_Hinge_Standalone.pdf in OLD/.

★ UPDATE NOTICE – 13 April 2026 ★

This version corrects the Landauer-crossing framing in §1 and §4. The earlier text placed the crossing at "steps 36–37." The refined picture identifies THREE distinct thresholds that cluster in the 35–37 zone:

$n \cdot G = \ln 2$	at $n \approx 35.11$	(Landauer erasure cost – the crossing)
$n \cdot G = 1/\sqrt{2}$	at $n \approx 35.82$	(geometric damping)
$n \cdot G = 1/(2 \ln 2)$	at $n \approx 36.54$	(Shannon bound)

The Landauer crossing proper is the FIRST threshold, at $n = \ln(2)/G$, which sits between steps 35 and 36. The entire three-threshold cluster is contained inside the ceiling span (steps 32–50), consistent with the partition given in §4.

The factor-2 hinge at step 50 is unchanged.

Prior published versions remain on OSF file storage, renamed with the OLD- prefix and a date for traceability.

ABSTRACT

The integer 50 appears independently in six structures spanning four domains of physics and mathematics, each derived or observed without reference to the others:

1. Atomic electron shell capacity at $n = 5$: $2 \cdot 5^2 = 50$
2. First partial quotient of the continued fraction of $G = 1 - K_{\text{AUD}}$
3. Tower step where accumulated cost reaches the ceiling: $50 \cdot G \sim K_{\text{AUD}}$
4. Square root of the first correction denominator in the gap scaling formula: $1/2500 = 1/50^2$

5. Nuclear magic number: 50 protons = Tin, the element with the most stable isotopes (Goeppert Mayer & Jensen, Nobel 1963)
6. DESI DR2 BAO cosmological distance ratios reproducing the $1/2500 = 1/50^2$ correction scale

where $K_{\text{AUD}} = \sqrt{2} * \ln(2) = 0.980258\dots$, the unique sub-unity value of $K(n) = \sqrt{n} * \ln(n)$ for integer $n \geq 2$.

These six appearances arise from six independent computations. None references the others. The convergence on $50 = 2 * 5^2$ is structural, governed by the H_4 primes {2, 5} that organize the framework.

1. THE FRAMEWORK IN BRIEF

$K_{\text{AUD}} = \sqrt{2} * \ln(2) = 0.980258143468547\dots$

The unique sub-unity value of $K(n) = \sqrt{n} * \ln(n)$ at integer $n \geq 2$. For all $n \geq 3$, $K(n) > 1$.

$G = 1 - K_{\text{AUD}} = 0.019741856531452\dots$

The irreducible gap. Approximately 2%. No binary system closes it.

The Binary Tower: 64 steps of accumulated cost $n * G$, with landmarks at steps 18 (Romeo), 32 (Floor), the 35–37 threshold zone (Landauer crossing at $n \approx 35.11$, geometric damping at $n \approx 35.82$, Shannon bound at $n \approx 36.54$), 50 (Ceiling), and 64 (Pivot).

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2. APPEARANCE 1 – ATOMIC ELECTRON SHELLS

SOURCE: Standard quantum mechanics (Schrodinger equation + Pauli exclusion principle, established since the 1920s).

The electron shell capacity formula:

$$\text{Capacity}(n) = 2n^2$$

At $n = 5$:

$$\text{Capacity}(5) = 2 * 5^2 = 50$$

This is the theoretical g-orbital threshold – the shell where 50 electrons would fill if atoms extended that far in practice.

The formula $2n^2$ contains two of K_{AUD} 's three ingredients explicitly: the factor 2 is BINARY (electron spin has exactly two states), and n^2 is QUADRATIC (angular momentum scales as n^2). The third ingredient, the Gaussian, enters through the electron wavefunctions themselves.

SHELL CAPACITIES AS TOWER LANDMARKS:

Shell n	Capacity	Tower Value	Framework Constant	Deviation
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n = 3	18	18G = 0.3554	$\sqrt{2}/4 = 0.3536$	+0.51%
n = 4	32	32G = 0.6317	$1/\phi = 0.6180$	+2.22%
n = 5	50	50G = 0.9871	K_AUD = 0.9803	+0.70%

The atom's shell capacities – 18, 32, and 50 – are the same numbers that appear as the tower's organizational landmarks. The shell capacities come from quantum mechanics; the tower landmarks come from $K_AUD = \sqrt{2} * \ln(2)$. They agree because both are built from the same three ingredients: $\sqrt{2}$, $\ln(2)$, and quadratic geometry.

3. APPEARANCE 2 – CONTINUED FRACTION OF THE GAP

SOURCE: Rational approximation theory (number theory, no physics input needed).

The continued fraction expansion of G:

$G = [0; 50, 1, 1, 1, 7, 1, 29, 2, 2, 1, \dots]$

The first non-zero partial quotient is exactly 50.

This means $1/50$ is the best single-fraction approximation to G:

$G \sim 1/50 = 0.02000$ (true value: 0.01974...)

Equivalently:

$K_AUD \sim 49/50 = 0.98000$ (true value: 0.98026...)

The rational approximation decomposes into framework primes:

$49/50 = 7^2 / (2 * 5^2)$
 $= \text{boundary}^2 / (\text{binary} * \text{pentagonal}^2)$

The continued fraction of K_AUD itself:

$K_AUD = [0; 1, 49, 1, 1, 1, 7, 1, 29, 2, 2, 1, \dots]$

Note the structural relationship: G's CF starts with 50, K_AUD's starts with 1 then 49. After these opening terms, both sequences are identical: 1, 1, 1, 7, 1, 29, 2, ...

The boundary prime 7 appears at position 6 in both CFs – recurring at depth, not just at the surface.

4. APPEARANCE 3 – BINARY TOWER CEILING

SOURCE: The tower cost function $n * G$, derived from K_AUD.

At step 50 of the Binary Tower:

$50 * G = 0.987093\dots$
 $K_AUD = 0.980258\dots$
 Overshoot: +0.697%

Step 50 is where the accumulated cost of binary organization reaches the ceiling – the value K_{AUD} itself. The tower "arrives home" at step 50: the cost of 50 gaps equals (approximately) the very constant that defines the gap.

This creates a self-referential lock: $G = 1 - K_{AUD}$ defines the gap, and $50 * G \sim K_{AUD}$ closes the circle. The number 50 is the step count at which the gap's own accumulation reconstructs its parent constant.

The tower partitions around step 50:

Steps 0-18: Romeo (first complete shell)
 Steps 18-32: Floor (golden ratio reciprocal)
 Steps 32-50: Ceiling (K_{AUD}) – contains the Landauer crossing
 Steps 50-64: Pivot ($\sqrt{\phi}$)

Partition: 18 - 14 - 18 - 14

The two 18-step spans and two 14-step spans mirror each other, forced by the Ramanujan-Nagell theorem ($2^5 - 5^2 = 7$). Step 50 sits at the second mirror point – the boundary between the ceiling span and the pivot span.

5. APPEARANCE 4 – GAP SCALING CORRECTION

SOURCE: The connection between G and the Feigenbaum constant δ , linking geometry to chaos theory.

The gap ratio:

$$\rho = \delta * G / (\delta - 14/3)$$

where $\delta = 4.6692\dots$ is Feigenbaum's universal period-doubling constant. This ratio decomposes as a correction series:

$$\rho = 400/11 - 1/2500 - 1/939939$$

$$\text{Error: } 4 * 10^{-14}.$$

The first correction term:

$$1/2500 = 1/50^2$$

$$2500 = 2^2 * 5^4 = (2 * 5^2)^2 = 50^2$$

The correction scale is the SQUARE of the hinge integer 50. Its prime factorization uses only H_4 primes {2, 5} – the same primes that compose 50 itself.

The correction hierarchy:

Order 0:	$400/11 = (2^4 * 5^2) / 11$	base term
Order 1:	$1/2500 = 1/50^2$	$\sim 10^{-4}$ (H_4 primes)
Order 2:	$1/939939 = 1/(3*7*11*13*313)$	$\sim 10^{-6}$ (full prime chain)

Each order introduces deeper framework primes. The first correction stays within {2, 5}. The second brings in {3, 7, 11, 13}. The hierarchy mirrors the prime entry sequence in the atom itself.

("Order" is used here as a scale-order label, distinct from the framework's evidence-level taxonomy in §11.)

6. THE NUCLEAR ECHO – MAGIC NUMBER 50

SOURCE: Nuclear shell model (Goeppert Mayer and Jensen, Nobel Prize 1963). Standard nuclear physics.

The nuclear magic numbers are: 2, 8, 20, 28, 50, 82, 126.

Nuclei with magic numbers of protons or neutrons have closed shells – maximum binding energy, maximum stability, minimum decay. They are the nuclear equivalent of noble gases in atomic physics.

50 protons = Tin (Sn).

Tin has 10 stable isotopes – more than any other element. This exceptional stability is a direct consequence of the closed proton shell at $Z = 50$. The nuclear magic number arises from the $1g_{9/2}$ orbital closure under the spin-orbit interaction, a mechanism entirely independent of the electron shell capacity formula $2n^2$ that also produces 50.

The same integer governs maximum stability in BOTH the electronic structure (shell 5 capacity) and the nuclear structure (5th magic number) of atoms, through different physics.

Furthermore: $82 = 50 + 32$ (ceiling step + floor step in the tower) is the atomic number of Lead – the heaviest quasi-stable element. Lead-208 is doubly magic (82 protons, 126 neutrons). At step 82 of the tower:

$82 * G = 1.61883\dots$
 $\phi = 1.61803\dots$
Deviation: 0.049%

This is the tightest match between any tower step and any framework constant. The sum of ceiling and floor steps equals the atomic number of the heaviest stable element, and its tower value matches the golden ratio to 0.049%.

7. THE COSMIC ECHO – DESI BAO AT THE 1/2500 SCALE

SOURCE: DESI Collaboration, Phys. Rev. D 112, 083514 (2025), Table IV. Published observational cosmology data.

The baryon acoustic oscillation (BAO) frozen ruler, imprinted at recombination, is measured across cosmic epochs by the DESI survey. Distance ratios between tracer populations reveal:

$DV(LRG2) / DV(BGS) = 2.020141$

The framework predicts:

$2 + G = 2.019742\dots$

Residual:

$$2.020141 - 2.019742 = +0.000399$$

$$1/2500 = 0.000400$$

$$\text{Match: residual} / (1/2500) = 0.998$$

The DESI isotropic distance ratio decomposes as:

$$\text{DV(LRG2)}/\text{DV(BGS)} = 2 + G + 1/2500 + 0(10^{-6})$$

The same $1/50^2$ correction scale from the gap scaling formula absorbs the cosmological residual to parts-per-million precision.

A second DESI pair shows the same scale:

$$\text{DM(QS0)}/\text{DM(LRG3+ELG1)} - \text{sqrt}(2) = +0.000406$$

$$1/\text{residual} \sim 2464 = 2^5 * 7 * 11$$

Both residuals cluster at $4.0 * 10^{-4}$ – the characteristic $1/2500$ correction scale predicted by the framework.

8. THE FACTORIZATION

$$50 = 2 * 5^2$$

This is the product of the two smallest H_4 primes – the primes that compose the order of the H_4 polytope symmetry group:

$$|H_4| = 14400 = 2^6 * 3^2 * 5^2$$

The factor 2 is the binary ingredient (information, spin, Landauer). The factor 5^2 is the pentagonal ingredient (golden ratio, angular momentum, icosahedral symmetry).

50 contains no other primes. It is the simplest non-trivial product of the H_4 foundation primes, elevated by one power of the pentagonal prime.

The correction $1/2500 = 1/50^2 = 1/(2^2 * 5^4)$ stays entirely within $\{2, 5\}$ – it is a pure H_4 -prime correction. Only when the second correction ($1/939939$) is needed do the higher framework primes $\{3, 7, 11, 13\}$ enter the series.

9. WHY 50 AND NOT ANOTHER NUMBER

The convergence on 50 is not arbitrary. Each appearance is forced:

ATOMIC: The shell capacity $2n^2$ at $n = 5$ gives 50. The formula is fixed by quantum mechanics. The only freedom is which shell number n we examine – and $n = 5$ is singled out because it is the shell whose tower value (50G) reaches the ceiling.

CONTINUED FRACTION: The CF of $G = [0; 50, \dots]$ is determined

entirely by the decimal expansion of $G = 1 - \sqrt{2} \ln(2)$.
The partial quotient 50 is a number-theoretic fact about the
transcendental number G .

TOWER: Step 50 reaches the ceiling because $50 * G \sim K_{\text{AUD}}$, which
is equivalent to $50 * (1 - K_{\text{AUD}}) \sim K_{\text{AUD}}$, or $K_{\text{AUD}} \sim 50/51$.
The fact that $K_{\text{AUD}} \sim 49/50$ (from the CF) and $K_{\text{AUD}} \sim 50/51$
(from the tower) are nearly the same approximation is itself a
consequence of K_{AUD} being close to 1.

GAP SCALING: The correction $1/2500 = 1/50^2$ emerges from the
continued fraction expansion of $\rho = \delta * G / (\delta - 14/3)$,
connecting geometry (G) to chaos (δ). The 4th convergent of
 ρ 's CF is $400/11$, and the first correction is $1/50^2$.

Correction (2026-09-11): $1/2500$ does not come from the continued
fraction. $\rho = [36; 2, 1, 3, 20, 2, 1, 384, \dots]$; its convergents
after $400/11$ are $8109/223$, ... – no 50, no 2500. Only $400/11$ is a
convergent; $1/2500$ is the rounded unit fraction chosen at the
second step of the gap scaling formula (Paper 4, §3.2). The
arithmetic $2500 = 50^2$ stands; the word "emerges" does not.

NUCLEAR: The magic number 50 arises from the spin-orbit splitting
of the $1g_{9/2}$ orbital in the nuclear potential well – physics
entirely independent of electron shells or K_{AUD} .

COSMIC: The $1/2500 = 1/50^2$ scale in DESI data is an observed
residual in published measurements, not derived from the framework.

Six routes to the same integer. No route references any other.

10. VERIFICATION

All values computed with mpmath at dps = 100 (verified twice):

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K_AUD = sqrt(2) * ln(2) = 0.980258143468547...
G = 1 - K_AUD = 0.019741856531452...
50 * G = 0.987092826572640... (overshoot +0.697% from K_AUD)
CF of G = [0; 50, 1, 1, 1, 7, 1, 29, 2, ...]
CF of K_AUD = [0; 1, 49, 1, 1, 1, 7, 1, 29, 2, ...]
49/50 = 0.98 (K_AUD - 49/50 = 0.000258...)
1/2500 = 0.0004
rho = 400/11 - 1/2500 - 1/939939 = 36.363235299...
82G = 1.61883... vs phi = 1.61803... (deviation 0.049%)
DV(LRG2)/DV(BGS) - (2 + G) = +0.000399 ~ 1/2500
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Sources:

Atomic shells: Standard QM textbooks (Schrodinger 1926, Pauli 1925)
Nuclear magic numbers: Goeppert Mayer & Jensen (Nobel 1963)
DESI BAO: arXiv:2503.14738, Table IV
Feigenbaum constant: $\delta = 4.669201609...$ (Feigenbaum 1978)

11. EVIDENCE CLASSIFICATION

LEVEL 1 – EXACT [Theorem]:

$50 = 2 * 5^2$ (electron shell capacity at $n = 5$)

CF of G begins with 50 (number-theoretic identity)
 $1/2500 = 1/50^2$ in gap scaling formula
50 is a nuclear magic number (standard physics)

LEVEL 2 – EMPIRICAL [Observation]:

DESI BAO residuals at $\sim 1/2500$ scale (published, measured)

LEVEL 3 – CONJECTURAL [Conjecture]:

$50G \sim K_{AUD}$ (tower near-miss, +0.70%)

$82 = 50 + 32 = Z$ of Lead, with $82G \sim \phi$ (0.049%)

These levels are not conflated. Each carries different evidential weight. The Level 1 items are mathematical facts. The Level 2 item is published observational data. The Level 3 items are patterns that derive significance from the convergence of the others – explicitly unproved and motivated by partial evidence. Labels follow the canonical methodology page; this is a definitional alignment and does not change the scope, honesty, or calculations of this document.

12. WHAT WOULD STRENGTHEN OR WEAKEN THIS

STRENGTHEN:

- DESI DR3/DR5 maintaining the $1/2500$ residual scale as statistical errors shrink ($\sim 30\%$ per data release)
- Independent BAO surveys (Euclid, Roman) showing the same correction scale in their distance ratios
- A derivation showing WHY the CF of G must begin with 50 (currently observed, not proven to be necessary)

WEAKEN:

- DESI DR3 showing the residual drifting away from $1/2500$
- A demonstration that the CF beginning with 50 is not structurally significant (i.e., that many transcendentals of this form have similar first partial quotients)
- Evidence that the nuclear magic number 50 and the electron shell capacity 50 are not independent (currently they arise from different physics, but a deeper connection could change the counting of independent appearances)

One integer. Six appearances. Four scales. Zero fitting.

The number $50 = 2 * 5^2$ organises electron filling, the fractional structure of G, the algebraic correction hierarchy, nuclear shell closure, and the frozen acoustic ruler across cosmic epochs.

It does this because it is the simplest non-trivial product of the H_4 foundation primes, and because K_{AUD} – the ceiling of binary information density – sits close enough to $49/50$ that the integer 50 becomes the natural quantum of approach.

D. B. – April 2026

Gap Geometry

<https://gap-geometry.github.io/sqrt2-ln2-geometric-constants-/about.html>

OSF: osf.io/zx4g7

K_AUD framework: November–December 2025
50 Hinge discovery (Grok, xAI): 17 March 2026
Documents prepared for publication: 3 April 2026
